

Answers to Referee's Questions

“Interpolating the Stochastic Heat and Wave Equations with Time-independent Noise: Solvability and Exact Asymptotics”

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We would like to thank the two anonymous reviewers for carefully reading our manuscript and for providing detailed reports, which have helped us to improve significantly the paper. We have revised our paper accordingly. Here are a list of changes that we have made. The page and equation numbers refer to the revised version of the manuscript.

1 Changes according to Reviewer 1

1. page 2: Fixed the "nonnegative and nonnegative-definite" typo
2. page 13: Fixed the typo involving $\mathcal{A}_{a,d}$.

2 Changes according to Reviewer 2

1. page 1, line 21, 23 and 24: Changed "exits" to "exists".
2. page 1, line 22: Added "as" to the sentence.
3. page 2, line 32: Added parenthesis to $f^{(m)}$.
4. page 2: We removed the mention of Z and Z^* and then reworded the last paragraph to accommodate this change.
5. page 4, line 48: Corrected the "Wiener" misspelling.
6. page 5: We corrected the condition on r so that $r < 1/2$.
7. page 5, line 15: We added a space after " $p \geq 2, \dots$ ".
8. page 6, line 23: Added the word "illustrate".

9. page 6, line 34: Changed "exits" to "exists".
10. page 7, item (b): We referenced the theorem and figure below.
11. page 7, line 7: Changed "to" to "with".
12. page 7, line 14: We removed the word "the".
13. page 7, line 37 and line 48: Changed "exits" to "exists".
14. page 7, line 41: Reworded the paragraph starting on this line.
15. page 10, line 44: Changed "Laplace" to "Laplacian".
16. page 10: We removed the repeated $\alpha = a = d = 2$ case from the bottom of the page.
17. page 13: Fixed the $\gamma(x - y)$ typo.
18. page 14, line 3: Changed "exits" to "exists".
19. page 15, line 11: Added a reference to the Skorohod integral at the end of this paragraph.
20. page 15, line 31: Reworded the sentence to say that the moment "diverges to infinity" for $t > T_2$.
21. page 15, line 39: Fixed the Wiener misspelling.
22. page 16, line 20: Fixed the order of the arguments for the kernel $\tilde{f}_n$.
23. page 16, line 38: Fixed the "Wiener" misspelling.
24. page 16, line 40: Changed R to $\mathbb{R}$.
25. page 17, line 23: Added the assumption that $\alpha < \min\{2a, d\}$.
26. page 17, Theorem 3.5: Added the assumption that $\alpha < \min\{2a, d\}$.
27. page 20: Replaced " f " with " b " in the definition for $||\cdot||_{q,w}$.
28. page 20, line 23: Removed the reference to [1, Lemma 1.6].
29. page 21, line 9: Added an equation link to where this is referenced.
30. page 21, Lemma 3.9: Added the assumption on α . Directed the reader to the definition of $\sigma(a, d, \alpha)$. Replaced "2" with " α " in the exponent.
31. page 21, Lemma 3.10: Made clear that Lemma 3.5 is from the current paper. Fixed the g typo inside the definition of Λ .

32. page 22, line 23: Replaced " $\mathbb{R}$ " with " $\mathbb{R}_+$ ".
33. page 22, line 33: Replaced " 2 " with " α " in the exponent.
34. page 22: All $\sigma(d, \alpha)$ were replaced with $\sigma(a, d, \alpha)$.
35. page 22: We reverted back to using γ instead of the Riesz kernel. ie equation (3.34).
36. page 22, Lemma 3.11: Replaced $\gamma(x - y)$ with $\gamma(x + y)$.
37. page 23, line 4: Corrected the centered equation.
38. page 23: Reverted back to using $\gamma(\cdot)$ instead of $|\cdot|^{-\alpha}$.
39. page 23, line 48: Replaced " f " with " g ".
40. page 24, line 3: Added reference to Lemma 3.11 for the use of σ .
41. page 24: Removed the excess periods, fixed the exponents by changing " 2 " to " α ", and also replaced " $\sigma(d, \alpha)$ " with " $\sigma(a, d, \alpha)$ ".
42. page 25, line 3: Made it clear we are only proving part I.
43. page 25, Eq 4.5: Fixed the $\xi_{\sigma(j)}$ error.
44. page 25, line 50: Gave a reference to the function K .
45. page 26, Eq 4.6: Explained the notation.
46. page 26 and 27, Eq 4.2: $\alpha < 2a$ was assumed in the statement of the theorem and we added a reference to clarify that. We also reordered the argument as suggested.
47. page 27, Eq 4.14: Replaced " n " with " 2 ".
48. page 27, line 28: Added a reference for the inequality.
49. page 27, line 35: Made a comment about the switch to $x = 0$.
50. page 28, Lemma 5.1: Relaxed the assumption on α and referenced ρ as was suggested.
51. page 28, line 59: Added a reference for H .
52. page 29: Fixed all the $+1$ typos.
53. page 29: Explained the derivation of C_n .
54. page 29, line 56: Fixed the misspelling of hypercontractivity.
55. page 30: You were correct, we made the change to r ie $r \in [0, 1/2)$.
56. page 30, line 32: Fixed the typo involving the reference to Lemma 5.1.

- 57. page 32, line 37: Added "it".
- 58. page 33, line 22: Fixed the subscript in $\mathbb{E}(U_n(t))$.
- 59. page 33, line 50: Replaced H with $\mathcal{H}$.
- 60. page 35: Fixed all g_h typos.
- 61. page 37, Eq 6.19: Fixed the n_t typo.
- 62. page 38, line 10: Fixed the t_p typo.
- 63. page 39: Reformatted the proof of (4.9) and took into account the suggestions made.
- 64. page 40: Removed the duplicated reference.

3 Additional minor corrections

In addition to the above changes suggested by the two referees, we implemented some minor corrections which are highlighted in blue in the manuscript. We list these below.

- 1.